Voting Intelligence Framework

Layered classification. Calibrated confidence.
Continuous learning, qualified for your corpus.

ISO 15022 and narrative notices / Customer-controlled processing

ONE CONTROLLED QUALIFICATION LIFECYCLE

Learn from reviewed examples. Validate independently.
Release only within a measured operating policy.

01	02	03	04	05	06	07
Frozen foundation	**Customer training**	**Validation selection**	**Model freeze**	**Confidence calibration**	**Independent qualification**	**Monitored release**

A framework that learns

Frozen foundations, customer-specific training and a repeatable path to release qualification.

Decisions that remain accountable

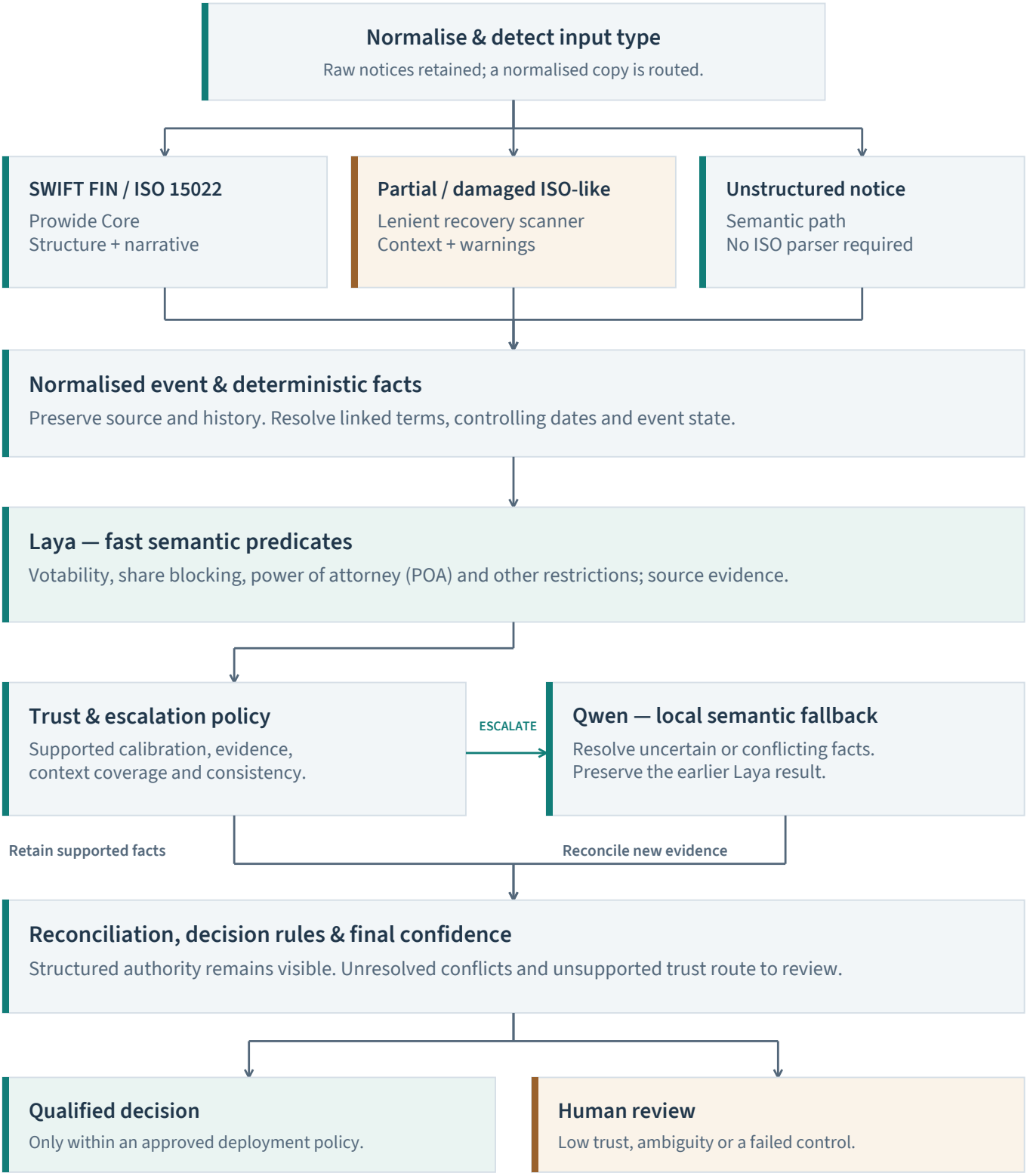
Source evidence, scoped confidence, human review and versioned learning from operational feedback.

Automation starts disabled. Customer-specific qualification and explicit deployment approval are required before automated processing.

01 / RUNTIME ARCHITECTURE

From notice to decision

A layered classification pipeline. Models and routing are qualified for each deployment.



Outputs: votable / not votable / restricted / deadline passed / review. Source, model, evidence and timing remain traceable. Parser success is not full conformance validation.

02 / EXECUTABLE CUSTOMER QUALIFICATION

Seven stages. Recorded evidence.

Each stage produces a durable artefact. A failed gate retains review-only operation.

	STAGE & ACTION	RECORDED OUTPUT
1	**Frozen foundation** Verify pinned weights, runtime, semantic criteria and source hashes. Provision model files separately from the application.	Foundation manifest Verified model identity
2	**Customer training** Import and review the corpus. Train Laya or a compatible local LLM using only the approved TRAIN role.	Dataset + job manifest Checkpoint + lineage
3	**Validation selection** Compare candidates on VALIDATION: critical recall, complete-output correctness, evidence, coverage and latency.	Selection receipt Candidate configuration
4	**Model freeze** Freeze weights, normalisation, prompts, criteria, rules and runtime. A change creates a new candidate identity.	Immutable model bundle Reload verification
5	**Confidence calibration** Fit and check calibration on separate groups. Select the operating policy on development data; seal the full release.	Calibrator + support report Frozen operating policy
6	**Independent qualification** Persist predictions before opening protected test answers. Check the fixed policy, critical recall and accepted-output precision.	Qualification receipt Passed / not met + reasons
7	**Monitored release** Apply independent approval and the deployment permission. Start under observation; retain review, audits and rollback.	Approved release identity Operational monitoring plan

Before independent testing: weights, prompt, preprocessing, rules, routing, calibrator and acceptance thresholds are fixed. Test outcomes never select these settings.

03 / DATA ISOLATION & RELEASE GATES

Independent data. Explicit permission.

Near-duplicates, linked messages and amendment chains remain in the same partition.

DATA ROLE	PERMITTED USE
TRAIN	Update model weights or adapters. Calibration and frozen-test answers are excluded.
VALIDATION	Select candidates, training settings and the operating policy. These results are development evidence.
CALIBRATION	Fit correctness estimates and check them on disjoint groups. Never use these labels to update model weights.
FROZEN TEST	Independently qualify the sealed release. Predictions are saved before protected answers are accessed.

Labels retain their provenance. AI-authored data qualifies engineering mechanics. Representative, independently expert-reviewed customer labels are required for customer performance and automation qualification.

THE DEPLOYMENT GATE CHECKS

Accepted-output precision

Supported lower bounds for the full requested output, not just average field accuracy.

Critical-positive recall

Explicit support and missed-positive counts for voting and material restrictions.

Measured review volume

Coverage and review rates meet the agreed operational policy.

Evidence & input scope

Supported language, complete context and trustworthy source evidence.

Installation qualification

The exact model, runtime and server pass capacity, recovery and load checks.

Independent promotion

Authorised approval, a current qualification and the deployment permission.

Any gate not met → review-only. Targets are customer-agreed business policies, not regulatory guarantees.

04 / CONFIDENCE & OPERATIONAL REVIEW

Trust is measured, not asserted.

Confidence is specific to the model, release configuration and declared customer population.

01 Native model score

A Laya class score or an available token-level model probability. Useful as an input signal; it is not automatically a probability of correctness.

02 Calibrated predicate confidence

A held-out estimate of whether a particular predicate prediction is correct within its supported population. Unsupported conditions retain unavailable confidence.

03 Final-decision confidence

A separately evaluated correctness estimate for the decision. It does not establish that every restriction is correct; all requested operational fields still need support.

A high score does not promise a correct answer.

Operating thresholds are chosen on development data, then checked independently. The customer sees measured coverage, errors and review volume with sample support and uncertainty bounds.

ROUTE TO REVIEW WHEN

Trust is insufficient

Low or unavailable calibration support; an unknown critical fact.

Evidence is unresolved

Conflicting current terms, weak support or an unverified quotation.

The input is outside scope

Unsupported language, incomplete context or a materially different input population.

An execution control fails

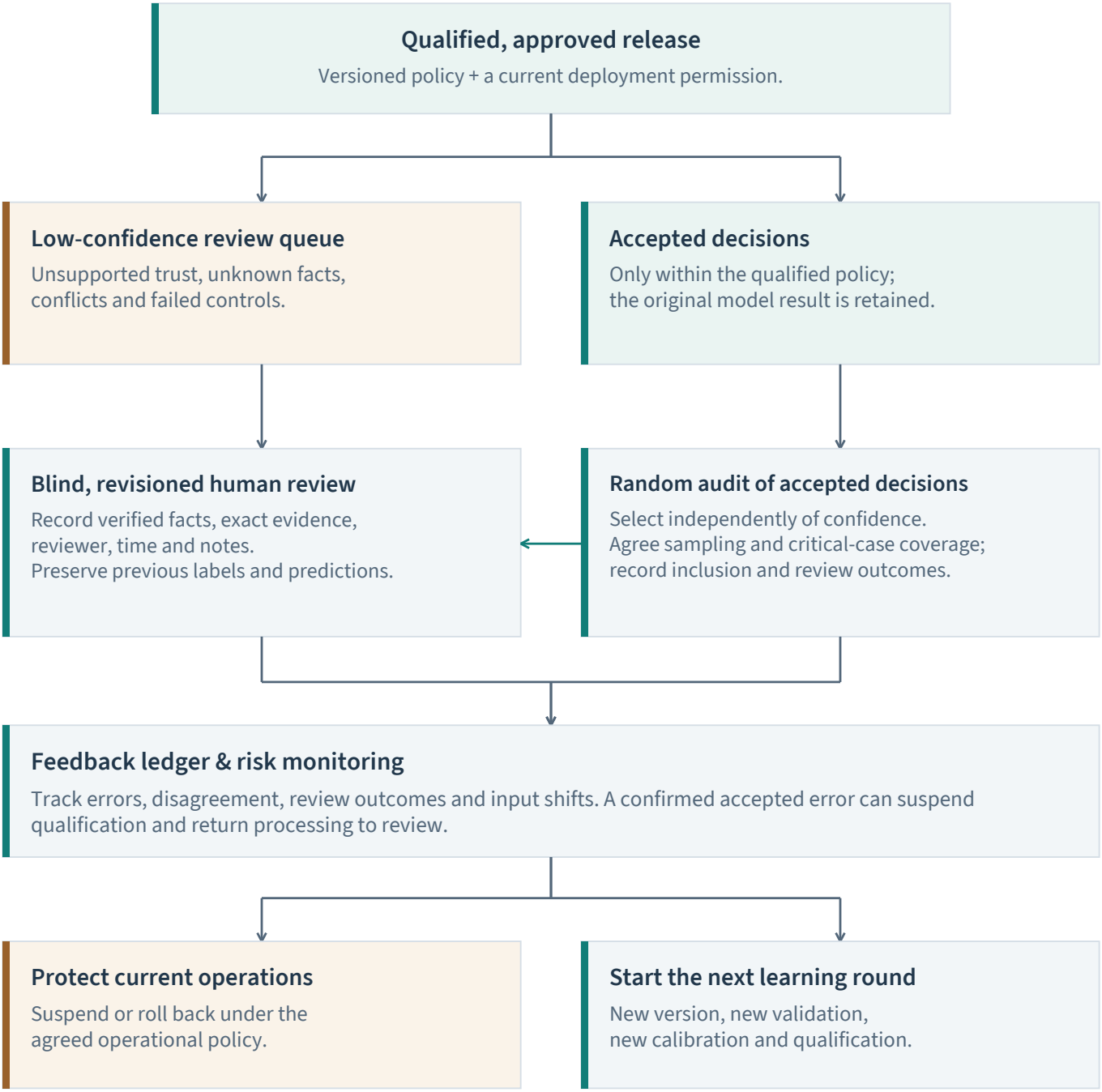
Schema failure, deterministic conflict or an unavailable qualified runtime.

No universal confidence threshold, guaranteed zero-error rate or fixed low review percentage is promised. High-confidence decisions remain subject to audit.

05 / MONITORED RELEASE & CONTINUOUS LEARNING

Review the uncertain. Audit the accepted.

A required operating model: targeted review and random audits provide complementary feedback.



No silent online learning. Operational corrections become versioned candidates. They do not alter active weights or frozen test labels in place.

Automated random-audit sampling is a planned control to implement and verify during acceptance. Existing review, revision history and error-triggered suspension provide the underlying feedback path.

06 / DEPLOYMENT BOUNDARY & CUSTOMER ACCEPTANCE

Your framework. Your data.

Single-tenant Linux container deployment in the customer environment; exact hardware is qualified separately.

Framework delivery

- Application and container recipes
- Prowide + lenient recovery adapter
- Laya / local-LLM interfaces
- Training, evaluation and calibration
- Review, evidence and audit workflows
- Pinned foundation installation manifests

Customer workspace

- Raw notices and event history
- Reviewed labels and protected splits
- Customer checkpoints / adapters
- Customer-specific calibrators
- Predictions and review revisions
- Qualification and operational receipts

Foundation weights are provisioned separately and verified by revision and file hash. Model inference requires no external generative API; installation downloads are a separate provisioning step.

EXCLUDED FROM THE SHIPPED PRODUCT

AI-authored qualification corpora, protected answer keys, test-trained models, test calibrators and research-run artefacts.

IMPLEMENTATION & ACCEPTANCE STATUS

Learning, checkpoint and review workflows	Implemented; controlled-data verification
Customer confidence and performance	Customer-specific qualification required
Exact Linux model / hardware profile	Installation qualification required
Random audits of accepted decisions	Automated sampling is a planned control
Automated operational classification	Disabled until deployment gate + approval

Customer onboarding begins with corpus/profile review, an agreed risk policy and a supported deployment profile.

This brief describes the architecture and qualification operating model. It is not an accuracy certificate or completed production acceptance. Customer accuracy remains unestablished until representative independent evaluation.